VADDI MOHITH RAJ

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CAREER OBJECTIVE

2nd-year B.Tech (AI/ML) student at SRMIST, Kattankulathur, applying for the Summer Internship 2026 on AI & ML in Electronic Cooling using ANSYS Icepak at the Center of Excellence for Electronic Cooling and CFD Simulation Lab. I bring production-grade ML expertise — including multi-algorithm comparative frameworks, full-stack AI pipelines, and SHAP explainability — and am eager to apply these skills to thermal management challenges in electronics, EV, and data center domains under the mentorship of Dr. B. K. Gnanavel.

EDUCATION

B.Tech — CSE (AI/ML) | SRMIST, Kattankulathur *2024 – 2028 (Expected)*
CGPA: 7.4 | Currently: 4th Semester

Intermediate (Class XII) | Excellencia Junior College, Hyderabad, Shamirpet
2022 – 2024
Percentage: 88%

TECHNICAL SKILLS

AI / ML	XGBoost, scikit-learn, SBERT (all-MiniLM-L6-v2), FAISS, TF-IDF, LightGBM, Neural Networks, RAG pipelines
Data Science	SHAP Explainability, Comparative Benchmarking (RF, LightGBM, SVM, NN), Feature Engineering, Statistical Analysis
LLM & Prompt Engineering	Claude API (Anthropic), OpenAI API, Groq (LLaMA-3.3), Chain-of-Thought, JSON output structuring
Full-Stack Dev	Next.js 14, React, FastAPI, Flask, MySQL 8.0, PostgreSQL, REST APIs
Deployment & Tools	Railway, Render, Vercel, Docker (basics), Git / GitHub, Pinecone, Google Colab
Languages	Python, JavaScript (ES6+), C, SQL

Domain Interest	Thermal Management, CFD Simulation, Electronic Cooling, EV/Data Center AI applications

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KEY PROJECTS

GeneRisk AI — DNA-Based Cancer Risk Prediction Platform (2024 – Present)
generisk-ai.vercel.app | Flask + Next.js + XGBoost | Deployed on Railway & Vercel

• Built a multi-algorithm ML framework (XGBoost, RF, LightGBM, SVM, Neural Networks) for cancer risk prediction — achieving 97.37% accuracy and ROC-AUC 0.9950 — mirroring the comparative benchmarking approach applicable to thermal simulation model selection in ANSYS Icepak.

• Integrated SHAP explainability to generate per-patient feature attribution, enabling interpretable AI decisions — directly relevant to validating AI-driven thermal predictions against CFD ground truth.

• Deployed ML API on Railway with Gunicorn and Three.js 3D frontend on Vercel; managed full production lifecycle from model training to user-facing dashboard.

CineGenome V3.0 — AI Film Recommendation Engine (2024 – Present)
Next.js 14 + FastAPI + MySQL 8.0 | Local ML: TF-IDF + SBERT + FAISS | Built by Stream Minds

• Engineered a fully local hybrid ML system (TF-IDF, Sentence-BERT embeddings, FAISS vector search) across 3,477 records — demonstrating ability to design and optimize ML pipelines for structured engineering datasets, analogous to thermal simulation metadata.

• MySQL 8.0 backend normalized to 3NF; resolved production-level ISP DNS-blocking by migrating cloud DB — demonstrating resilience engineering applicable to simulation data pipelines.

Google Colab — AI/ML Research Notebooks (2024 – Present)
Digital Image Processing & Data Science | Academic & Self-directed Research

• Developed DIP Colab notebooks covering ROI extraction, texture feature analysis, and image compression — foundational techniques applicable to thermal imaging and heat-map analysis in electronic cooling systems.

• Completed Scaler Advanced DBMS course and Forage Gen AI Data Analytics simulation — strengthening analytical pipeline skills relevant to processing ANSYS Icepak simulation outputs with ML models.

HACKATHONS & ACHIEVEMENTS

• AppXcelerate 1.0 (SRMIST KTR, 2024): Built ResumeAI — Groq API (LLaMA-3.3) + 3D UI + three-strategy JSON parsing pipeline.
• Pentathon 3.0 / Webathon 3.0 (2024): Finalist — competitive web development track with full AI integration.
• FASTATHON 2025 / QwikInnovate / OssomeHacks: Submitted Guardian Grid, an AI women's safety platform with ORACLE (XGBoost + LSTM), PHANTOM, and JUSTICE engines.

CERTIFICATIONS

• Python Using AI — Be10x (2024)
• Forage Gen AI Powered Data Analytics Job Simulation
• Scaler Advanced DBMS Course

WHY THIS INTERNSHIP

The Center of Excellence for Electronic Cooling and CFD Simulation Lab's focus on applying AI & ML to thermal management in electronics, EV, and data center systems is a direct intersection of my production ML expertise and my academic AI/ML curriculum. I am eager to work under Dr. B. K. Gnanavel to learn ANSYS Icepak workflows, contribute ML-based surrogate models for thermal prediction, and translate simulation data into interpretable AI outputs — gaining enterprise-grade research experience that will accelerate my goal of building impactful AI systems.